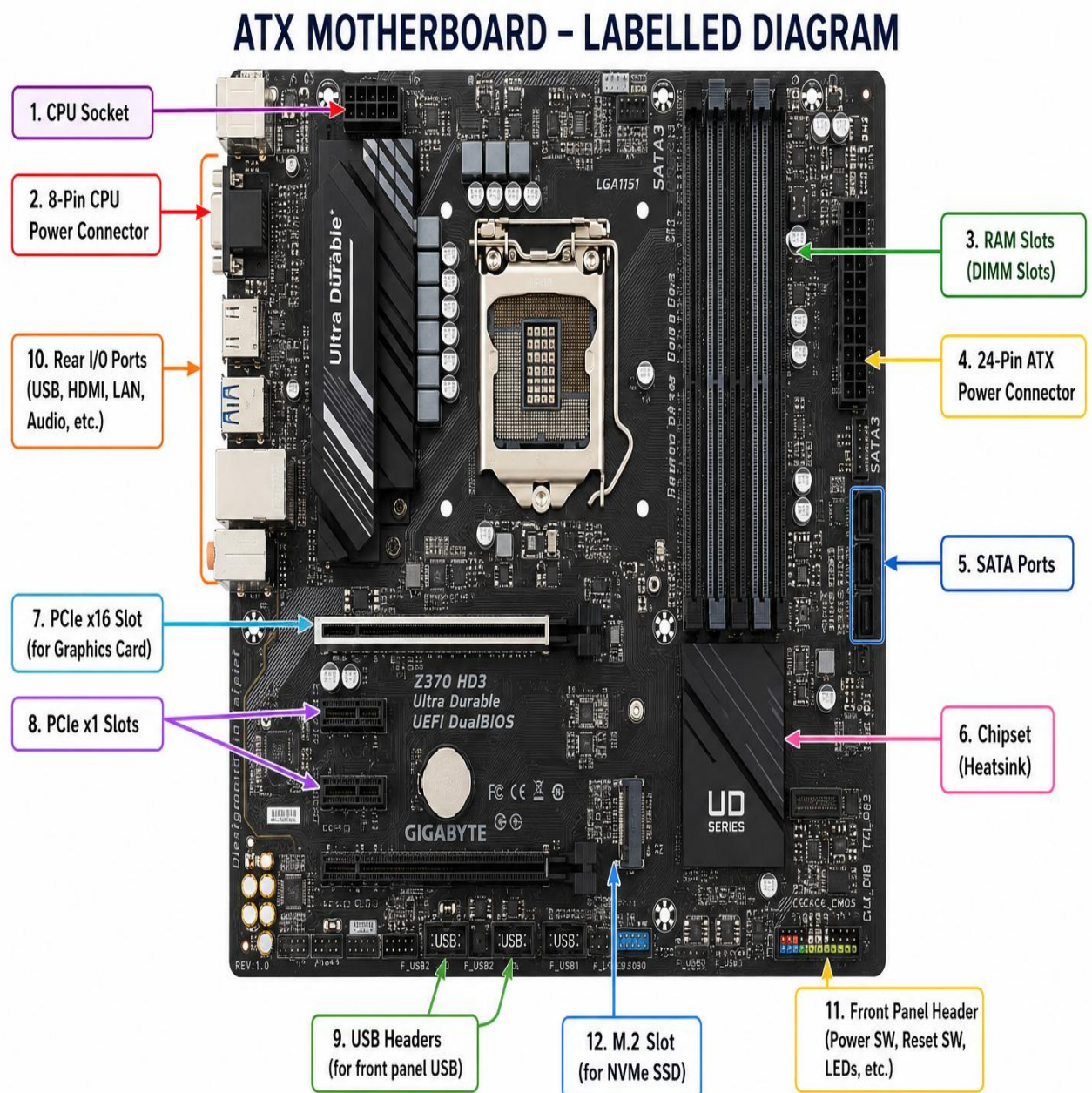


HARDWARE

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.)



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

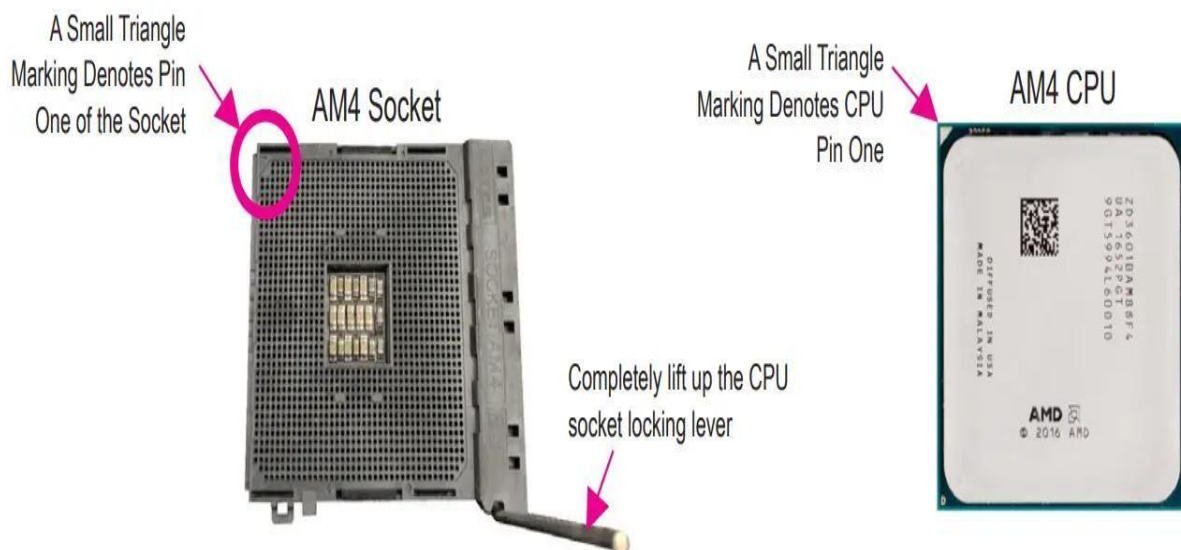
Popular CPU Socket Types with Compatible CPUs

1. LGA1151 (Intel)



- Brand: **Intel**
- **Compatible CPU**
Example: Intel Core i7-9700K
- **Used in:** 8th & 9th generation Intel processors

2. AM4 (AMD)



- **Brand:** AMD
- **Compatible CPU Example:** AMD Ryzen 5 5600X
- **Used in:** Ryzen 1000 to Ryzen 5000 series

3. LGA1200 (Intel)



- **Brand:** Intel
- **Compatible CPU Example:** Intel Core i5-10400
- **Used in:** 10th & 11th generation Intel processors

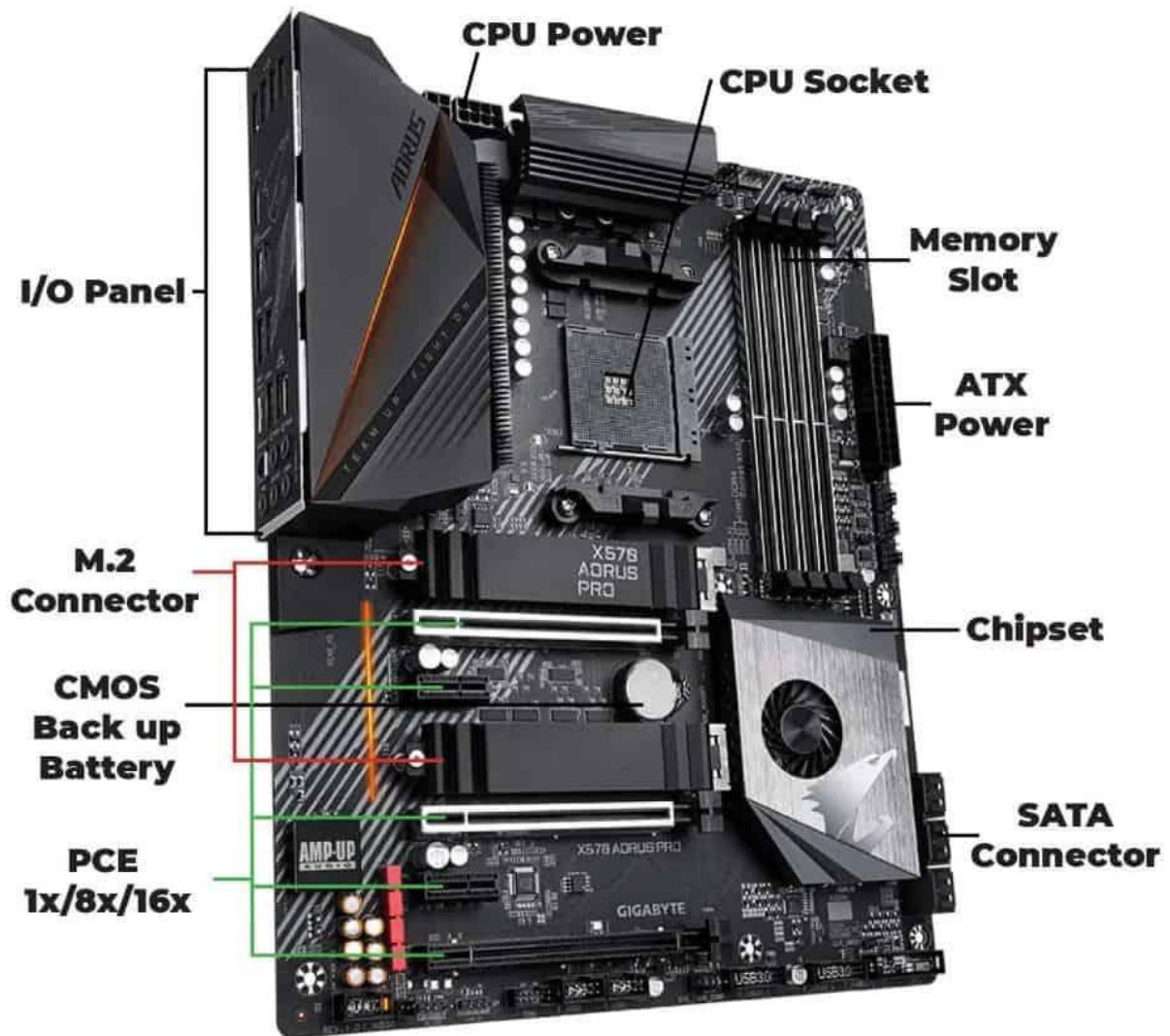
Three popular CPU socket types used in modern computers are LGA1151, AM4, and LGA1200. LGA1151 is a socket developed by Intel and is commonly used for 8th and 9th generation Intel processors; one compatible CPU model for this socket is the Intel Core i7-9700K. Another widely used socket is AM4, which is designed by AMD and supports a large range of Ryzen processors from the 1000 to the 5000 series; for example, the AMD Ryzen 5 5600X is compatible with the AM4 socket. Lastly, LGA1200 is also an Intel socket that supports 10th and 11th generation processors, and a suitable CPU for this socket is the Intel Core i5-10400. These socket types are important because they determine which processors can be installed on a motherboard, ensuring proper compatibility and performance.

3. Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

The CPU (Central Processing Unit) is the main part of the computer that performs all calculations and executes instructions. It acts like the brain, handling tasks such as opening applications, processing data, and controlling other components. The chipset, on the other hand, is a group of electronic components on the motherboard that manages communication between the CPU and other hardware parts like RAM, storage devices, and input/output ports.

When you open a music app like Spotify, both the CPU and chipset work together to complete the task smoothly. First, when you click on the app, the CPU processes the command and starts executing the program. At the same time, the chipset helps transfer data from the storage device (SSD or HDD) to the RAM so the application can load quickly. The CPU then continues to run the app, manage playback, and process user actions like play, pause, or skip. Meanwhile, the chipset ensures proper communication between the CPU, audio hardware, and other connected devices. In this way, the CPU does the main processing work, while the chipset supports and coordinates the data flow between different components.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each on it.



- **CPU Socket** – The square area in the center where the processor is installed.
- **RAM Slots (DIMM Slots)** – Long vertical slots near the CPU used to install memory.
- **PCIe Slot** – Long horizontal slot used for graphics cards and expansion cards.
- **SATA Ports** – Small connectors (usually on the side) used to connect HDD/SSD.

- **24-pin Power Connector** – Large connector on the edge of the motherboard for main power supply.